

sPHENIX isolated photons on the world x_T -scaling curve

PPG12 — isolated photon cross section in $p+p$ at $\sqrt{s} = 200$ GeV

2026-06-02

Abstract

We place the PPG12 sPHENIX isolated-photon measurement on the world x_T -scaling curve of direct-photon cross sections, reproducing Fig. 1 (left) of arXiv:1901.10950 (Bock, Hard Probes 2018). After scaling by $(\sqrt{s})^{4.5}$, the sPHENIX points fall on the universal curve and lie at or just below the PHENIX 200 GeV direct-photon data, consistent within combined uncertainties. This is an external, cross-experiment check on the absolute normalization of the PPG12 cross section, spanning $\sqrt{s} = 31.6$ GeV to 13 TeV.

1 What Figure 1 is, and why the overlay is meaningful

x_T scaling exploits the leading-order pQCD expectation that the invariant cross section for a hard $2 \rightarrow 2$ process factorizes as $E d^3\sigma/dp^3 = (1/\sqrt{s})^n F(x_T)$, with n governed by the effective parton-parton scattering and the PDF/fragmentation evolution. Multiplying every measurement by $(\sqrt{s})^n$ collapses the different beam energies onto a single curve $F(x_T)$. The PHENIX 200 GeV direct-photon paper (PRD86 072008) made exactly this figure, and the Bock proceedings is a later, broader rendering of the same compilation. Because the compilation already contains a 200 GeV entry (PHENIX), adding the PPG12 200 GeV point is a direct apples-to-apples comparison at fixed $\sqrt{s}$, plus a test that our spectral shape tracks $F(x_T)$.

PPG12 measures *isolated* photons ($E_T^{\text{iso}}, R = 0.3 < 4$ GeV truth definition). The compilation mixes isolated (LHC) and inclusive-direct (PHENIX, E706) measurements, as the figure caption signals with the parenthetical “(isolated) direct photon”. To make the same-energy comparison explicit we plot PHENIX both as published (inclusive-direct) and multiplied by its own measured isolated/inclusive-direct ratio (PRD86 Fig. 13a). PPG12 sits at or just below the lower (isolated) of the two PHENIX curves.

2 Data provenance

HEPData is the natural source, but its download endpoints are behind a Cloudflare managed challenge that cannot be solved from the analysis cluster. The numbers HEPData serves are digitized from the published papers, and arXiv e-print sources are not blocked, so we pulled the identical data tables straight from the papers’ own L^AT_EX sources. Each external extraction was performed by one agent and independently re-extracted and checked point-by-point by a second.

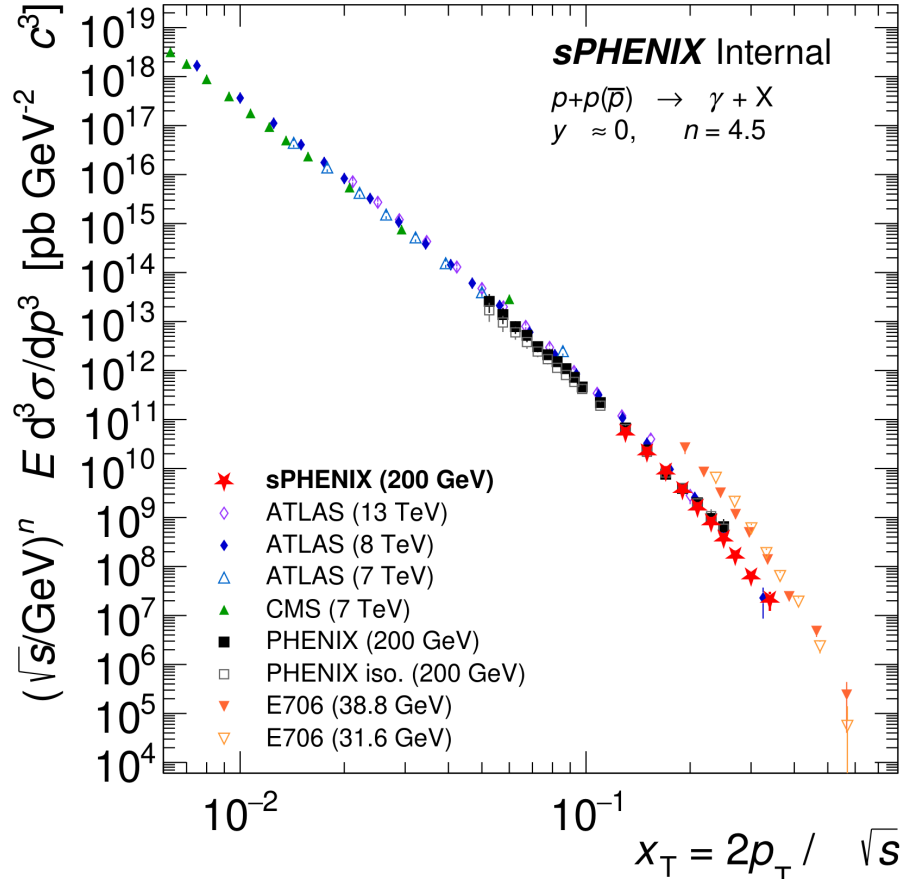


Figure 1: $(\sqrt{s}/\text{GeV})^n E d^3\sigma/dp^3$ (units $\text{pb GeV}^{-2} \text{ c}^3$, $n = 4.5$) versus $x_T = 2p_T/\sqrt{s}$ at $y \approx 0$ for $p + p(\bar{p}) \rightarrow \gamma + X$. The PPG12 sPHENIX measurement (red stars) is overlaid on the world compilation.

Experiment	$\sqrt{s}$ [GeV]	Reference	observable	η range	iso	pts
sPHENIX (this)	200	PPG12 Photon_final	$d\sigma/dE_T$	$ \eta < 0.7$	yes	10
PHENIX	200	PRD86 072008	$E d^3\sigma/dp^3$	$ \eta < 0.25$	no	18
PHENIX (isolated)	200	PRD86 Fig.10×13a	$E d^3\sigma/dp^3 \times R$	$ \eta < 0.25$	der.	18
CMS	7000	PRL106 082001	$d\sigma/dE_T$	$ \eta < 1.45$	yes	11
ATLAS	7000	PLB706 150	$d\sigma/dE_T$	$ \eta < 0.6$	yes	8
ATLAS	8000	JHEP08(2016)005	$d\sigma/dE_T$	$ \eta < 0.6$	yes	22
ATLAS	13000	PLB770 473	$d\sigma/dE_T$	$ \eta < 0.6$	yes	14
E706	38.8	PRD73 032004	$E d^3\sigma/dp^3$	$-1.0 < y < 0.5$	no	9
E706	31.6	hep-ex/0407011	$E d^3\sigma/dp^3$	$ y < 0.75$	no	8

The pre-arXiv SppS, ISR, and fixed-target entries that complete the Bock figure (CDF and D0 at 1.8 TeV, UA1/UA2, the ISR experiments, E704, NA24, WA70, UA6, and the two older PHENIX papers) live only on HEPData and were not pulled here. They are expected to sit on the same band (they do in the original Bock and PHENIX figures) and do not change the comparison.

3 Conversion to the common observable

Each spectrum is reduced to the invariant cross section at midrapidity and then x_T -scaled. For an invariant cross section (PHENIX, E706), $I \equiv E d^3\sigma/dp^3$ is used as published. For an η -integrated differential cross section (ATLAS, CMS, PPG12),

$$I = \frac{1}{\Delta\eta} \frac{d\sigma}{dE_T} \frac{1}{2\pi p_T},$$

with $\Delta\eta$ the width of the central η bin used (1.2 for ATLAS $|\eta| < 0.6$, 2.9 for CMS $|\eta| < 1.45$, 1.4 for PPG12 $|\eta| < 0.7$). Cross-section units are normalized to pb. Then $Y = (\sqrt{s})^{4.5} I$ and $x_T = 2p_T/\sqrt{s}$.

The strongest validation is internal. The three ATLAS measurements (7, 8, 13 TeV) are independent yet agree to $\lesssim 40\%$ at every shared x_T , and the PHENIX and PPG12 200 GeV spectra agree to $1.2\times$. CMS sits up to $\sim 2\times$ above the ATLAS curves in the overlap, consistent with centroid bias in its very wide E_T bins. A wrong $\Delta\eta$, 2π , or unit factor in any one dataset would be far larger than these residuals.

4 Findings

1. The sPHENIX points lie on the universal x_T curve over the reported range $x_T \approx 0.13$ – 0.34 ($E_T = 12$ – 36 GeV). The $E_T < 12$ GeV trigger-turn-on bins and the overflow bin are excluded, matching the range PPG12 reports.
2. **Same energy:** at $x_T = 0.13$, PPG12/PHENIX(incl.-direct) = 0.82 and PPG12/PHENIX(isolated) = 0.87. The deficit is expected (isolation removes a fragmentation component) and is inside the combined uncertainties (PPG12 ≈ 14 – 21% , PHENIX ≈ 16 – 20% in this overlap region).
3. **Cross-energy** scaling holds at the factor- ~ 2 level (LHC-to-RHIC spread $2.15\times$ at $x_T = 0.13$; PHENIX–PPG12 $1.22\times$). Part of this spread is the isolated-vs-inclusive-direct mixing in the compilation, not pure x_T -scaling violation.
4. E706 (31.6, 38.8 GeV) sits ~ 7 – $8\times$ above the band at high x_T — the long-known fixed-target direct-photon excess, reproduced here and not a conversion artifact.

The headline for the analysis: the PPG12 absolute normalization is consistent with the world data once placed on the x_T -scaling curve, and in particular with the only other midrapidity direct-photon measurement at 200 GeV.

5 Verification

The numerical re-derivation agent recomputed every quoted number from the ROOT file and raw tables from scratch: PPG12 ($E_T = 13, 25$ GeV from `h_unfold_sub_result`), PHENIX, and ATLAS conversions all reproduce to floating-point (MATCH, 0.0000%). The figure passed the plot-cosmetics review. The “within a factor of 2” at $x_T = 0.13$ is literally $2.15\times$ when the highest LHC point is compared to the lowest RHIC point, so the statement is meant in the round sense.

Reproduce

```
cd plotting
python3 xtscaling/convert.py
root -l -b -q 'xtscaling/plot_xtscaling.C'
```

Code and data live under `plotting/xtscaling/` (`convert.py`, `plot_xtscaling.C`, `data/`, `converted/`); the figure is `plotting/figures/xtscaling_compilation.pdf`.